ECON 2050 – Mathematical Economics

Basic Real Analysis III

Differentiability: Functions of one variable

Recall the definition of differentiability for functions of one variable:

Definition

Let $D \subseteq \mathbb{R}$ be open. The function $f : D \to \mathbb{R}$ is differentiable at $x \in D$ if there exists $f'(x) \in \mathbb{R}$ such that

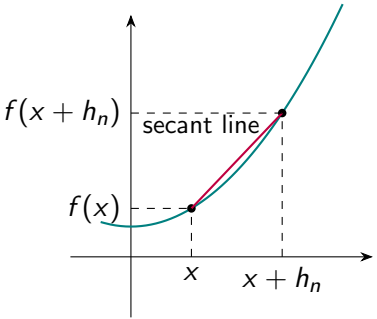
$$f'(x) = \lim_{n\to\infty} \frac{f(x + h_n) - f(x)}{h_n}$$

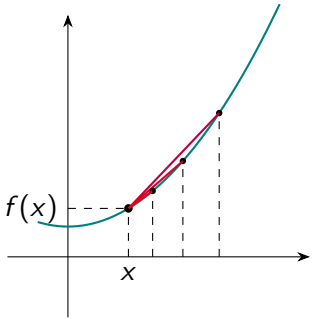
for all sequences (h_n) with $h_n \to 0$ and $h_n \neq 0$ for all n. In this case, $f'(x)$ is called the **derivative** of f at x.

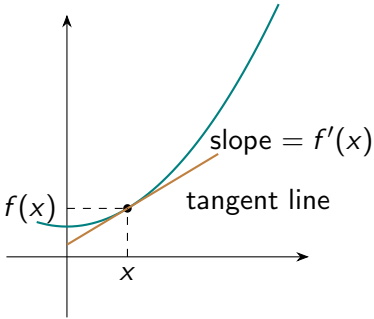
Equivalently, $f : D \to \mathbb{R}$ is differentiable at x_0 if and only if there exists $a \in \mathbb{R}$ such that

$$\lim_{n\to\infty} \frac{f(x_0 + h_n) - f(x_0) - ah_n}{h_n} = 0$$

for all sequences (h_n) with $h_n \to 0$ and $h_n \neq 0$ for all n. In this case, $a = f'(x_0)$.


$f(x+h_n)$
secant line
$f(x)$
x
$x+h_n$


$f(x)$
x


slope $= f'(x)$
tangent line
$f(x)$
x

Below are a list of results that are treated as well understood from previous courses in calculus and high school

$f(x)$	$f'(x)$
k	0
kx^n	knx^{n-1}
$\ln x$	$1/x$
e^x	e^x
$\cos x$	$-\sin x$
$\sin x$	$\cos x$

where $k \in \mathbb{R}$ is a constant.

The following results may also be familiar. These will be very useful throughout the course.

Theorem (Rules of differentiation)

Let $D \subseteq \mathbb{R}$ be open, and let $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}$ be differentiable at $x \in D$. Then $f + g$, fg, and f/g are differentiable at x, and

(1) $[f(x) + g(x)]' = f'(x) + g'(x)$;

(2) $[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$; *and*

(3) *if* $g(x) \neq 0$,

$$\left[\frac{f(x)}{g(x)}\right]' = \frac{g(x)f'(x) - g'(x)f(x)}{[g(x)]^2}.$$

These exercises are provided to "refresh" your calculus.

Exercises:

- Calculate the derivative of $h(x) = x \ln x$.

$$h'(x) = \ln x + x\left(\frac{1}{x}\right) = \ln x + 1.$$

- Calculate the derivative of $\hat{h}(x) = e^x/x$.

$$\hat{h}'(x) = \frac{e^x x - e^x}{x^2} = e^x \frac{x-1}{x^2}.$$

Theorem (Chain rule)

Let $D \subseteq \mathbb{R}$ be open, and let $f : D \to \mathbb{R}$ be differentiable at $x \in D$. Let $g : E \to \mathbb{R}$, where $f(D) \subseteq E$, be differentiable at $f(x)$, then the composition $g \circ f$ is differentiable at x, and

$$[(g \circ f)(x)]' = g'(f(x))f'(x).$$

The chain rule is often summarized as "the derivative of the outside times the derivative of the inside," but one must remember that the derivative of the outside function is evaluated at the inside function's *value* at x.

Exercises:

- Calculate the derivative of $\tilde{h}(x) = \ln(x^2 + 4)$.

  We first observe that $\tilde{h}(x) = (\tilde{g} \circ \tilde{f})(x)$, where $\tilde{g}(y) \equiv \ln y$, and $\tilde{f}(x) \equiv x^2 + 4$. By the chain rule,

$$\tilde{h}'(x) = \tilde{g}'(\tilde{f}(x))\tilde{f}'(x) = [\ln(\tilde{f}(x))]'\tilde{f}'(x) = \frac{1}{\tilde{f}(x)}\tilde{f}'(x) = \frac{2x}{x^2 + 4}.$$

- Calculate the derivative of $\bar{h}(x) = e^{1/x}$.

  We first observe that $\bar{h}(x) = (\bar{g} \circ \bar{f})(x)$, where $\bar{g}(y) \equiv e^y$, and $\bar{f}(x) \equiv 1/x$. By the chain rule,

$$\bar{h}'(x) = \bar{g}'(\bar{f}(x))\bar{f}'(x) = \left[e^{\tilde{f}(x)}\right]'\tilde{f}'(x) = e^{\tilde{f}(x)}\tilde{f}'(x) = -\frac{e^{\frac{1}{x}}}{x^2}.$$

Differentiability: Functions of several variables

In the remainder of today's lecture, we generalize the idea of differentiation to functions of several variables.

(Total) differentiability

Definition

Let $D \subseteq \mathbb{R}^k$ be open. The function $f : D \to \mathbb{R}$ is **(totally) differentiable at** $\boldsymbol{x} \in D$ if there exists $\boldsymbol{a} \in \mathbb{R}^k$ such that

$$\lim_{n\to\infty} \frac{|f(\boldsymbol{x}+\boldsymbol{h}_n) - f(\boldsymbol{x}) - \boldsymbol{a}\cdot\boldsymbol{h}_n|}{\|\boldsymbol{h}_n\|} = 0$$

for all sequences $(\boldsymbol{h}_n)$ with $\boldsymbol{h}_n \to \boldsymbol{0}$ and $\boldsymbol{h}_n \neq \boldsymbol{0}$ for all n. In this case, we call $Df(\boldsymbol{x}) \equiv \boldsymbol{a}$ the **derivative** of f at $\boldsymbol{x}$.

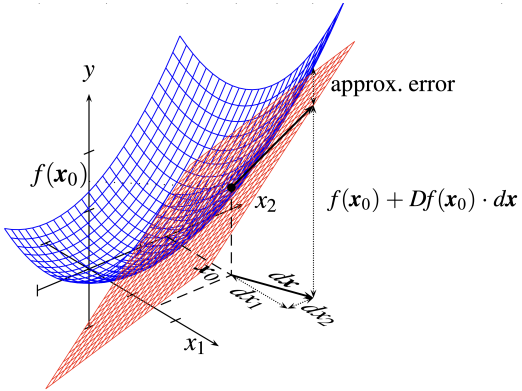
If f is differentiable at $\boldsymbol{x}$ for all $\boldsymbol{x} \in D$, then we say that f is **differentiable**. If in addition, the "derivative function" $\boldsymbol{x} \mapsto Df(\boldsymbol{x})$ is continuous, then we call f **continuously differentiable**.

The definition can be a bit abstract, but the geometric intuition should be straightforward, which is a natural generalization of the differentiability concept for functions of one variable.

- If $k = 1$,
  - $Df(x_0) \in \mathbb{R}$ is a number and
  - g is a tangent line of f at x_0.
- If $k = 2$,
  - $Df(\boldsymbol{x}_0) \in \mathbb{R}^2$ is a pair of numbers and
  - g is a tangent plane of f at $\boldsymbol{x}_0$.
- If $k = n$,
  - $Df(\boldsymbol{x}_0) \in \mathbb{R}^n$ is a vector of numbers and
  - g is a tangent hyperplane of f at $\boldsymbol{x}_0$.
- Once we know the tangent hyperplane g to f at $\boldsymbol{x}_0$, we can use it as an approximation of f in the neighborhood of $\boldsymbol{x}_0$.

Example:

- In blue, the graph of $f : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto \frac{1}{3}(x_1^2 + x_2^2)$.
- In red, the graph of $g : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto f(1.5, 1.5) + (1, 1) \cdot (x_1 - 1.5, x_2 - 1.5)$.



g is tangent to f at $\boldsymbol{x}_0 = (1.5, 1.5)$, and we claim that $Df(1.5, 1.5) =$ (1,1) .

- The (total) derivative of f allows us to approximate f "in any direction."
- A second type of derivative of f just asks how $y = f(\boldsymbol{x})$ changes if we move in a particular direction from $\boldsymbol{x}_0$.
- Such an idea is usually captured by the concept of "directional derivative"
- For now, we consider a special case, where we only move in a single dimension/element, say, x_i

Partial derivatives

Definition

Let $f : D \to \mathbb{R}$, with $D \subseteq \mathbb{R}^k$ open. If for all sequences (h_n) with $h_n \to 0$ and $h_n \neq 0$ for all n,

$$\lim_{n\to\infty} \frac{f(x_1, ..., x_i + h_n, ..., x_k) - f(x_1, ..., x_i, ..., x_k)}{h_n}$$

exists and equals to the same finite real number, then we denote this limit by $\frac{\partial f(\boldsymbol{x})}{\partial x_i}$, and say that f is **partially differentiable at $\boldsymbol{x}$ with respect to** x_i. In this case, $\frac{\partial f(\boldsymbol{x})}{\partial x_i}$ is called the **partial derivative** of f at $\boldsymbol{x}$ with respect to x_i.

If f is partially differentiable at $\boldsymbol{x}$ with respect to all x_i, then we call it **partially differentiable at $\boldsymbol{x}$**. If If f is partially differentiable at all $\boldsymbol{x} \in D$, we call it **partially differentiable**. If, in addition, all partial derivatives of f are continuous, then we call f **continuously partially differentiable.**

- For example, if they exist, the partial derivatives of a function $f : \mathbb{R}^2 \to \mathbb{R}$ are

$$\frac{\partial f(x,y)}{\partial x} = \lim_{n\to\infty} \frac{f(x+h_n,y) - f(x,y)}{h_n}$$
$$\frac{\partial f(x,y)}{\partial y} = \lim_{n\to\infty} \frac{f(x,y+h_n) - f(x,y)}{h_n}$$

- Like calculating derivatives of functions of one variables, we can sometimes calculate partial derivatives directly from the definition, that is, by directly determining corresponding the limit
- However, it is much easier to apply the methods for differentiation of a single variable to this setting. That is, we differentiate with respect to the variable of interest whilst "holding all others constant".
- Sometimes we wish to to calculate the partial derivatives at a point of interest; to that end, we first calculate the partial derivative functions and evaluate it at the point.

Example: The partial derivatives of $f : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto \frac{1}{3}(x_1^2 + x_2^2)$ are

$$\frac{\partial f(x_1, x_2)}{\partial x_1} = \frac{2}{3}x_1 \qquad \frac{\partial f(x_1, x_2)}{\partial x_2} = \frac{2}{3}x_2$$

- The partial derivatives at $(x_1, x_2) = (1.5, 1.5)$ are:

$$\frac{\partial f(1.5, 1.5)}{\partial x_1} = \frac{\partial f(1.5, 1.5)}{\partial x_2} = \frac{2}{3} \cdot 1.5 = 1$$

- Similar to a derivative of a function of one variable, the partial derivative $\frac{\partial f(\boldsymbol{x}_0)}{\partial x_i}$ represents the instantaneous rate of change f at $\boldsymbol{x}_0$ with respect x_i.
- $\frac{\partial f(1.5,1.5)}{\partial x_1} = 1$ signifies that the value of f at point $(1.5, 1.5)$ increases by one unit for a unit increase in x_1 *while holding* x_2 *fixed*.

Exercises:

- Calculate the partial derivatives of $g(x, y) = x^2 + xy + y + 1$.

$$\frac{\partial g(x, y)}{\partial x} = 2x + y, \qquad \frac{\partial g(x, y)}{\partial y} = x + 1.$$

- Calculate the partial derivatives of $h(x, y) = e^{xy} + y \ln x + 3$.

$$\frac{\partial h(x, y)}{\partial x} = ye^{xy} + \frac{y}{x}, \qquad \frac{\partial h(x, y)}{\partial y} = xe^{xy} + \ln x.$$

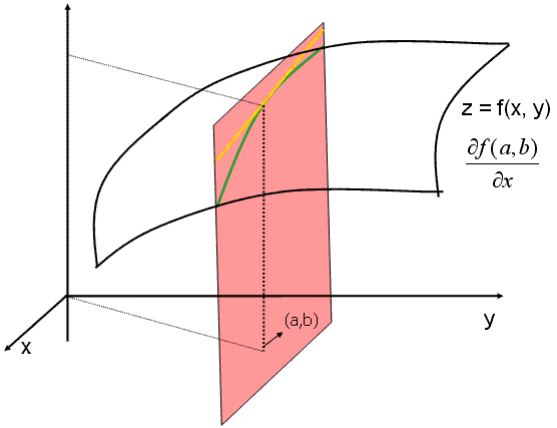
Some partial derivatives in Economics

Sometimes, partial derivatives are so important that economists give them special names:

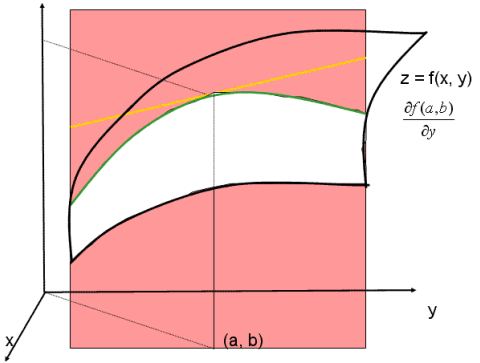
- Consider a production function $F : \mathbb{R}^2_{++} \to \mathbb{R}, (K, L) \mapsto Q(K, L)$.
  - $\frac{\partial F}{\partial K}$ is called the marginal product of capital.
  - $\frac{\partial F}{\partial L}$ is called the marginal product of labor.
- Consider a utility function $u : \mathbb{R}^k_{++} \to \mathbb{R}, \boldsymbol{x} \mapsto u(\boldsymbol{x})$.
  - $\frac{\partial u}{\partial x_i}$ is called the marginal utility of good i.

Geometric interpretation of partial derivatives

We can interpret the graph of $x \mapsto f(x, b)$ (green) as a curve at the intersection of the graph of $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto f(x, y)$ and the plane $\{(x, y, z) \in \mathbb{R}^3 : y = b\}$ (red), and $\frac{\partial f(a,b)}{\partial x}$ as the slope of the tangent line (yellow) to this curve at $(a, b, f(a, b))$.



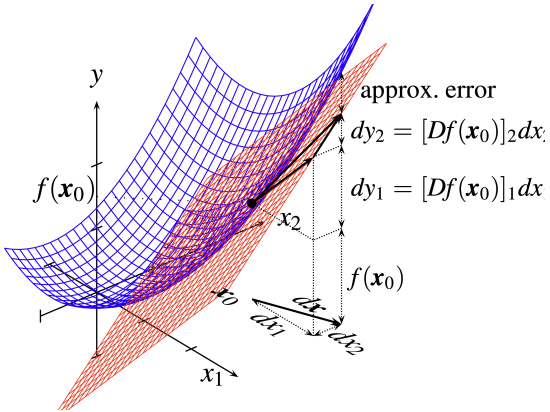
We can interpret the graph of $y \mapsto f(a, y)$ (green) as a curve at the intersection of the graph of $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto f(x, y)$ and the plane $\{(x, y, z) \in \mathbb{R}^3 : x = a\}$ (red), and $\frac{\partial f(a,b)}{\partial y}$ as the slope of the tangent line (yellow) to this curve at $(a, b, f(a, b))$.



Relation between partial and total derivatives

One way to derive $[Df(\boldsymbol{x}_0)]_1$ and $[Df(\boldsymbol{x}_0)]_2$ analytically is by exploiting the fact that g is affine and we can thus "decompose" $\mathrm{d}y = g(\boldsymbol{x}_0 + \mathrm{d}\boldsymbol{x}) - g(\boldsymbol{x}_0)$:

$$\mathrm{d}y = \mathrm{d}y_1 + \mathrm{d}y_2 = [Df(1.5, 1.5)]_1 \mathrm{d}x_1 + [Df(1.5, 1.5)]_2 \mathrm{d}x_2 = 1 \cdot \mathrm{d}x_1 + 1 \cdot \mathrm{d}x_2$$



Let $D \subseteq \mathbb{R}^k$ be open. We define the **gradient** $\nabla f(\boldsymbol{x}_0)$ of f at $\boldsymbol{x}_0$ as the vector whose coordinates are the partial derivatives of f:

$$\nabla f(\boldsymbol{x}_0) = \left(\frac{\partial f(\boldsymbol{x}_0)}{\partial x_1}, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_2}, \ldots, \frac{\partial f(\boldsymbol{x}_0)}{\partial x_k} \right)$$

Example: Consider $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto xy$.

- The gradient of f at (x, y) is $\nabla f(x, y) = (y, x)$.
- The gradient of f at $(3, -1)$ is $\nabla f(3, -1) = (-1, 3)$.

Theorem

Let $D \subseteq \mathbb{R}^k$ be open. If $f : D \to \mathbb{R}$ is differentiable (i.e., has a total derivative) at $\boldsymbol{x}_0$, then the total derivative of f at $\boldsymbol{x}_0$ is equal to the gradient of f at $\boldsymbol{x}_0$

$$Df(\boldsymbol{x}_0) = \nabla f(\boldsymbol{x}_0)$$

Tangent hyperplanes

We now can also determine tangent hyperplanes and estimate changes to values of differentiable functions f:

- Suppose $D \subseteq \mathbb{R}^k$ is open and $f : D \to \mathbb{R}$ differentiable at $\boldsymbol{x}_0$. Then $Df(\boldsymbol{x}_0) = \nabla f(\boldsymbol{x}_0)$ and the tangent hyperplane to f at $\boldsymbol{x}_0$ is an affine function

$$g : \mathbb{R}^k \to \mathbb{R}, \boldsymbol{x} \mapsto f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)(\boldsymbol{x} - \boldsymbol{x}_0)$$

- Writing $\Delta y \equiv f(\boldsymbol{x}_0 + \mathrm{d}\boldsymbol{x}) - f(\boldsymbol{x}_0)$ and $\mathrm{d}y \equiv g(\boldsymbol{x}_0 + \mathrm{d}\boldsymbol{x}) - g(\boldsymbol{x}_0)$ we obtain

$$\Delta y \approx \mathrm{d}y = \nabla f(\boldsymbol{x}_0) \cdot \mathrm{d}\boldsymbol{x} = \frac{\partial f(\boldsymbol{x}_0)}{\partial x_1}\mathrm{d}x_1 + \ldots + \frac{\partial f(\boldsymbol{x}_0)}{\partial x_k}\mathrm{d}x_k$$

  For $\mathrm{d}\boldsymbol{x}$ sufficiently close to $\mathbf{0}$, the change in the tangent hyperplane approximates the change in f.

(First-order) Taylor approximation

- When $\boldsymbol{x} - \boldsymbol{x}_0$ sufficiently small, $f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)(\boldsymbol{x} - \boldsymbol{x}_0)$ can be used to *approximate* $f(\boldsymbol{x})$:

$$f(\boldsymbol{x}) = f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)(\boldsymbol{x} - \boldsymbol{x}_0) + O(\boldsymbol{x}, \boldsymbol{x}_0)$$

  where $O(\boldsymbol{x}, \boldsymbol{x}_0)$ becomes negligible when $\boldsymbol{x}$ is sufficiently close to $\boldsymbol{x}_0$.

- Therefore, for $\boldsymbol{x}$ close to $\boldsymbol{x}_0$,

$$f(\boldsymbol{x}) \approx f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)(\boldsymbol{x} - \boldsymbol{x}_0)$$

  this is called the (first-order) **Taylor approximation** of f at the point $\boldsymbol{x}_0$.

(Total) differentiability vs. partial differentiability

Caution: In $\mathbb{R}^1$, by definition, f is differentiable in x_0 if and only if f is partially differentiable in x_0. This is *not* true in $\mathbb{R}^k$ if $k \geq 2$.

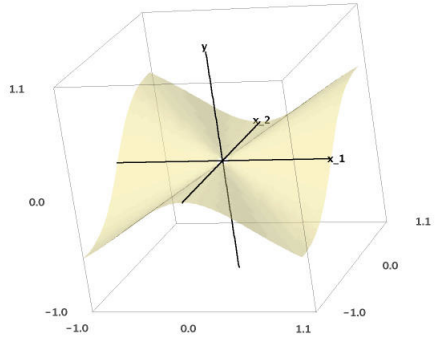
- If $D \subseteq \mathbb{R}^k$ is open and $f : D \to \mathbb{R}$ is differentiable at $\boldsymbol{x}_0 \in D$, then all partial derivatives exist at $\boldsymbol{x}_0$. In other words, f is partially differentiable at $\boldsymbol{x}_0$.
- However, the converse is not true: If f is partially differentiable at $\boldsymbol{x}_0 \in D$, f is *not necessarily* (totally) differentiable at $\boldsymbol{x}_0$.

Example:

- All partial derivatives of

$$f : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto \begin{cases} \frac{x_1^2 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq (0, 0) \\ 0 & \text{if } (x_1, x_2) = (0, 0) \end{cases}$$

at the point $(0, 0)$ exist, but f is not differentiable at $(0, 0)$.



If it exists, the derivative of f at $(0,0)$ is given by (a_1, a_2) such that:

$$\lim_{n\to\infty} \frac{|f(x_{1_n}, x_{2_n}) - f(0,0) - a_1 x_{1_n} - a_2 x_{2_n}|}{\sqrt{x_{1_n}^2 + x_{2_n}^2}} = 0$$

or,

$$\lim_{n\to\infty} \left| \frac{x_1^2 x_2}{x_{1_n}^2 + x_{2_n}^2} - a_1 x_{1_n} - a_2 x_{2_n} \right| \cdot \frac{1}{\sqrt{x_{1_n}^2 + x_{2_n}^2}} = 0$$

1. Take the sequence $(x_{1_n}, x_{2_n}) = \left(\frac{1}{n}, \frac{1}{n}\right)$. Then the above limit becomes:

$$\lim_{n\to\infty} \left| \frac{\frac{1}{n^3}}{\frac{2}{n^2}} - \frac{a_1}{n} - \frac{a_2}{n} \right| \frac{1}{\frac{\sqrt{2}}{n}} = \lim_{n\to\infty} \left| \frac{1}{2\sqrt{2}} - \frac{a_1 + a_2}{\sqrt{2}} \right| = 0$$

which is satisfied if and only if

$$a_1 + a_2 = \frac{1}{2} \tag{1}$$

2. Take the sequence $(x_{1_n}, x_{2_n}) = \left(\frac{1}{n^2}, \frac{1}{n}\right)$. Then the above limit becomes:

$$\lim_{n\to\infty} \left| \frac{\frac{1}{n^5}}{\frac{1}{n^4} + \frac{1}{n^2}} - \frac{a_1}{n^2} - \frac{a_2}{n} \right| \frac{1}{\sqrt{\frac{1}{n^4} + \frac{1}{n^2}}} = \lim_{n\to\infty} \left| \frac{1}{n + n^3} - \frac{a_1}{n^2} - \frac{a_2}{n} \right| \frac{n^2}{\sqrt{1 + n^2}} = 0$$

which is satisfied if and only if

$$a_2 = 0 \tag{2}$$

3. Take the sequence $(x_{1_n}, x_{2_n}) = \left(\frac{1}{n}, \frac{1}{n^2}\right)$. Then the above limit becomes:

$$\lim_{n\to\infty} \left| \frac{\frac{1}{n^4}}{\frac{1}{n^2}+\frac{1}{n^4}} - \frac{a_1}{n} - \frac{a_2}{n^2} \right| \frac{1}{\sqrt{\frac{1}{n^2}+\frac{1}{n^4}}} = \lim_{n\to\infty} \left| \frac{1}{n^2+1} - \frac{a_1}{n} - \frac{a_2}{n^2} \right| \frac{n^2}{\sqrt{n^2+1}} = 0$$

which is satisfied if and only if

$$a_1 = 0 \tag{3}$$

Since there is no a_1, a_2 satisfying all equations (1)-(3), the function f is not differentiable at $(0, 0)$.

We now check that all partial derivatives exist. Recall the definition of partial derivative with respect to x_1 at $\boldsymbol{x}_0$:

$$\lim_{n\to\infty} \frac{f(x_{1_0} + h_n, x_{2_0}) - f(x_{1_0}, x_{2_0})}{h_n}.$$

For $\boldsymbol{x}_0 = (0, 0)$, this simplifies to

$$\lim_{n\to\infty} \frac{f(h_n, 0) - f(0, 0)}{h_n};$$

and the given f this becomes (recalling that the sequence (h_n) is such that $h_n \to 0$ and $h_n \neq 0$ for all n)

$$\lim_{n\to\infty} \frac{\frac{(h_n)^2 \cdot 0}{(h_n)^2 + 0^2} - 0}{h_n} = 0.$$

An analogous argument shows that the definition of partial derivative with respect to x_2 at $\boldsymbol{x}_0$ also exists and equals zero.

So we have to be careful: Blindly computing the partial derivatives and assuming that the corresponding tangent hyperplane approximates f well can backfire!

However:

Theorem

If $D \subseteq \mathbb{R}^k$ is open and $f : D \to \mathbb{R}$ is continuously partially differentiable, then f is continuously differentiable.

Also note that if f has a tangent hyperplane at a point, then f is continuous at that point:

Theorem

If $D \subseteq \mathbb{R}^k$ is open and $f : D \to \mathbb{R}$ is differentiable at a point $\boldsymbol{x}_0 \in D$, then f is continuous at $\boldsymbol{x}_0$.